

Program: Diploma in Renewable Energy	
Course Code: 6292D	Course Title: Hybrid Electric Vehicles
Semester: 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L: 4 T: 0 P: 0)	Periods per semester: 60

Course Objectives:

- To impart basic knowledge of Hybrid Vehicles
- To understand the fundamental concept of components used in hybrid vehicles.
- To familiarize with battery parameters of Hybrid Vehicles.

Course Outcomes:

On completion of the course, the students will be able to:-

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Discover the drive trains and the evolution of a hybrid vehicle.	14	Applying
CO2	Summarize the components, modes of operation and energy saving potential of a hybrid vehicle.	15	Understanding
CO3	Outline the power electronic devices used in hybrid vehicles and its speed control methods.	16	Understanding
CO4	Summarize the parameters of batteries / energy storage and its subsystem	13	Understanding
	Series Test	2	

CO – PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3				3		
CO3	3				3		
CO4	3				3		

3- Strongly mapped**2- Moderately mapped****1- Weakly mapped**

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Discover the drive trains and the evolution of a hybrid vehicle.		
M1.01	Explain the introduction and evolution of hybrid electric vehicles.	3	Understanding
M1.02	Outline the performance of a hybrid vehicle.	3	Understanding
M1.03	Summarize the basic concepts of hybrid traction and drive trains.	4	Understanding
M1.04	Examine the fuel efficiency of a hybrid vehicle.	4	Applying
Introduction – Hybrid Vehicle Introduction-History of Evolution of Hybrid Vehicle Characteristics & Performance -Basics of vehicle performance(definitions only), Ideal performance of a vehicle power plant Basics of Traction & Topologies - Basic concept of hybrid traction, introduction to various hybrid drive-train used in hybrid vehicle(basic concept only) Efficiency of fuel -Fuel efficiency analysis.			
CO2	Summarize the components, modes of operation and energy saving potential of a hybrid vehicle.		
M2.01	Explain the components and design considerations of a hybrid vehicle.	4	Understanding

M2.02	Illustrate the drivetrain configurations and requirement of electric motor	4	Understanding
M2.03	Explain the modes of operation of hybrid electric vehicle	4	Understanding
M2.04	Compare electric vehicle with conventional vehicle in terms of energy saving potential	3	Understanding
	Series Test – I	1	
Introduction of electric components and design considerations -Introduction to electric components used in hybrid and electric vehicles, Basic concept of electric traction, design of power control strategies in hybrid electric vehicle(considerations only) Drive train configurations & electric motor -Different drive train configurations(block diagram only),Requirement of electric motors used in electric vehicle Modes of operation & comparison -Operating modes of Electric motor dominated system, comparison of electric vehicle with petrol engine vehicles Energy saving potential of Hybrid drive trains - conceptual advantages of a hybrid over a conventional vehicle in terms of overall energy.			
CO3	Outline the power electronic devices used in hybrid vehicles and its speed control methods.		
M3.01	Explain the power electronics components and its application in drivetrains.	4	Understanding
M3.02	Understand the different motor speed control methods.	3	Understanding
M3.03	Identify the modes of operation of the dc motor in variable speed operation.	4	Understanding
M3.04	Explain the basic concept of drives.	5	Understanding
Power electronic components in hybrid vehicle -Major power electronic components used in hybrid vehicles(brief explanation only), Advantages and applications of Electric Drives. Speed control methods -Speed control methods in electric hybrid vehicles-flux control and armature control method Modes of operation of dc motor in variable speed operation –motoring, Regenerative braking, dynamic braking, plugging(brief explanation only) Configuration of drives - configuration of dc and induction motor drives(basic concept only)			

CO4	Summarize the parameters of batteries / energy storage and its subsystems.		
M4.01	Understand the requirements of energy storage in hybrid and electric vehicles.	3	Understanding
M4.02	Identify the different energy storage options.	4	Understanding
M4.03	Explain the basic parameters of Batteries.	3	Understanding
M4.04	Understand the basic concept of communication systems in hybrid vehicles.	3	Understanding
	Series Test – II	1	
Introduction to Energy Storage- Requirements in Hybrid and Electric Vehicles. Different Storage options- Batteries, Fuel Cell, Supercapacitor. Batteries - Basics – Types, Parameters – Capacity, Discharge rate, State of charge, state of Discharge, Depth of Discharge, Technical characteristics, Battery pack Design, Properties of Batteries. Communication and supporting subsystems- In Vehicle Network-Communication network(Basic concept only)			

Text / Reference

T/R	Book Title/Author
T1	Ehsani, M. Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press
T2	B.L.Theraja, A K Theraja: A textbook of Electrical Technology-Volume III- S.Chand Publications.
R1	Husain, I. Electric and Hybrid Electric Vehicles, CRC Press
R2	A.K. Babu, Electric & Hybrid Vehicles, Khanna Publishing House, New Delhi (Ed. 2018)
R3	Fuhs, A. E. Hybrid Vehicles and the Future of Personal Transportation, CRC Press,

R4	Gianfranco, Electric and Hybrid Vehicles: Power Sources, Models, Sustainability, Infrastructure And The Market, Pistoia Consultant, Rome, Italy,
R5	Chan C. C. and K. T. Chau, Modern Electric Vehicle Technology, Oxford Science Publication,
R6	Lechner G. and H. Naunheimer, Automotive Transmissions: Fundamentals, Selection, Design and Application, Springer
R7	Rashid, M. H. Power Electronics: Circuits, Devices and Applications, 3rd edition, Pearson,
R8	Moorthi, V. R. Power Electronics: Devices, Circuits and Industrial Applications, Oxford University Press

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/108/103/108103009/
2	https://onlinecourses.nptel.ac.in/noc20_ee99/preview
3	https://fame2.heavyindustries.gov.in/
4	http://www.anert.gov.in/
5	https://powermin.gov.in/